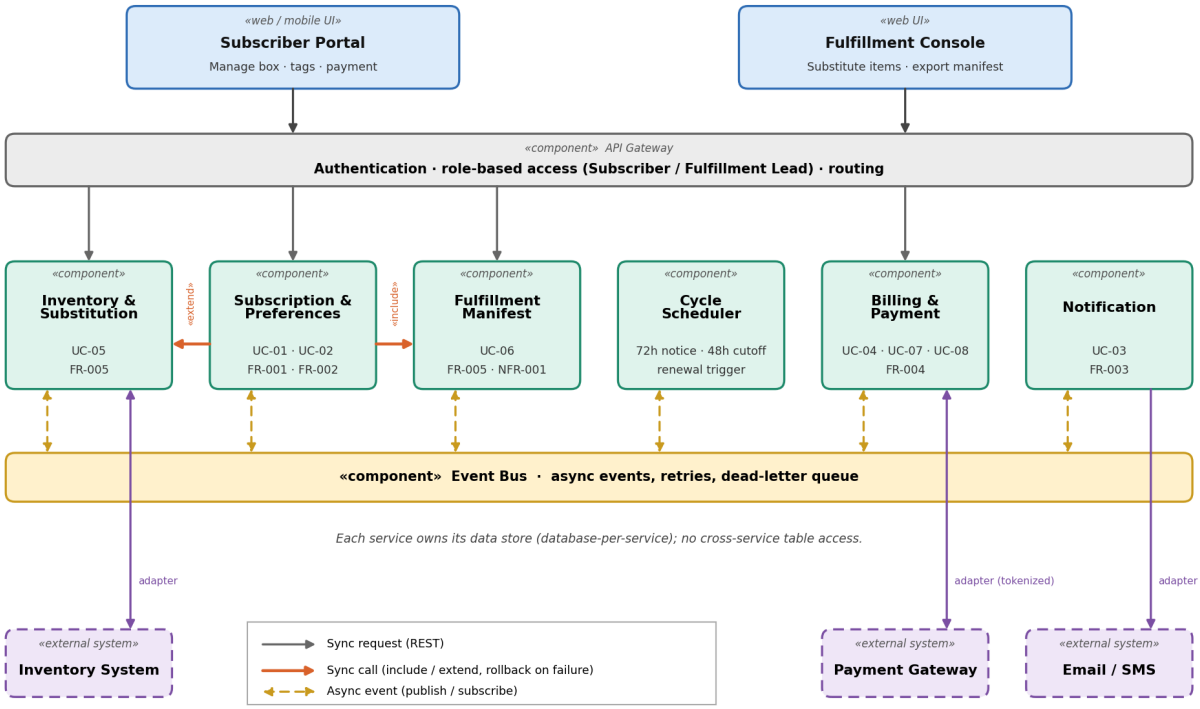
Architecture Justification

Problem Statement #35 — Customizable Subscription Box Scheduler

Chosen Style

Layered, event-driven services behind an API gateway. Two role-specific UIs (Subscriber, Fulfillment Lead) reach six domain services through one authenticated entry point. Services call each other synchronously only where consistency is required (UC-01 → UC-06 include, UC-05 extend) and communicate asynchronously via an event bus elsewhere. External systems sit behind adapters.



Key Decisions

Decision	Justification	Drives
Gateway + role-based auth	Single enforcement point for two actors with different permissions; keeps services free of auth logic.	UC-01–08
One service per domain	Maps 1:1 to the use-case model. Manifest can scale independently of Subscription during batch runs.	FR-001–005, NFR-001
Cycle Scheduler owns all time rules	One source of truth for the 72h notice, 48h cutoff and renewal trigger. Prevents rule drift across services.	FR-001, FR-003
Sync calls for include / extend	Swap/pause must commit only if the manifest refresh succeeds; failure triggers rollback (alt. flow 6a). Keeps profile and manifest consistent.	FR-001, UC-01
Async event bus	Notifications, payment retries and recommendation updates run off-path with retries and dead-letter handling, so they never block customization.	FR-002, FR-003, UC-08
Adapters for external systems	Isolates Payment Gateway, Inventory System and Email/SMS vendors. Only payment tokens are stored, limiting PCI scope.	FR-004, FR-005
Sharded batch manifest export	Partition 10,000 boxes across parallel workers and stream label output to meet the <1 min target.	NFR-001
Redundant stateless services	≥2 replicas across zones, replicated databases, deploy freeze during the 48h window. 99.5% of 48h allows ~14.4 min downtime.	NFR-002

Trade-offs

• More operational overhead than a monolith; mitigated by deploying as a modular monolith first and splitting along the same boundaries.
• Eventual consistency for notifications and billing retries; acceptable because neither blocks a subscriber action.
• Sync coupling between Subscription and Manifest is deliberate, trading some availability coupling for guaranteed consistency.